

VDA9.4

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Exercise 4 (Mapping Between Index and World Space, 5 Points)

Assume a uniform grid with origin at $\mathbf{p} = (1, 1, 2)^\top$, axis directions $\mathbf{e} = \frac{1}{\sqrt{2}}(1, 0, 1)^\top$, $\mathbf{f} = (0, 1, 0)^\top$, $\mathbf{g} = \frac{1}{\sqrt{2}}(1, 0, 1)^\top$, and axis spacings $dx = 1$, $dy = 2$, and $dz = 1$.

Hint: You can use a computer (e.g., Python) to do the computations in this task. It is acceptable if you specify rounded results.

a) Specify the affine transformation matrix from index space coordinates (i, j, k) to world space coordinates (x, y, z) . Then specify the corresponding mapping for transferring world space coordinates (x, y, z) back to index space coordinates (i, j, k) . (3P)

b) Use the transformation matrix to determine the world space position of the voxel with index coordinates $(2, 1, 5)$, and the displacement vector in world space that corresponds to incrementing the first index space coordinate by one voxel, the second one by two voxels, and the third one by minus one voxel. Which of the answers changes if we move the origin $\mathbf{p}$ and why? (2P)

a)

The transformation must convert the unit i , j , and k -vectors into $dx \cdot \mathbf{e}$, $dy \cdot \mathbf{f}$, $dz \cdot \mathbf{g}$ respectively. We deduce therefore that the linear part of the transformation

must be $\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 2 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} \end{pmatrix}$. The transformation must also transform the origin to

the point $\mathbf{p}$. Therefore the affine part must be $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, and thus the matrix of

the affine transformation from (i, j, k) to (x, y, z) is $\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 1 \\ 0 & 2 & 0 & -1 \\ \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

We can compute the transformation from (x, y, z) to (i, j, k) computing the

inverse matrix, which is $\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & \frac{-3}{\sqrt{2}} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

b)

The world (that is (x, y, z)) position of the voxel with index $(i, j, k) = (2, 1, 5)$, can be computed using the affine transformation matrix. As it is a point, its

affine component must be 1. As $\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 1 \\ 0 & 2 & 0 & -1 \\ \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 5 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{7}{\sqrt{2}} + 1 \\ 1 \\ \frac{-3}{\sqrt{2}} + 2 \\ 1 \end{pmatrix}$, the

(x, y, z) position of the point is $(\frac{7}{\sqrt{2}} + 1, 1, \frac{-3}{\sqrt{2}} + 2)$.

If we want to compute the world coordinates of the voxel space translation vector $(i, j, k) = (1, 2, -1)$, we can do so again using the matrix, but considering that as it is a vector, it must have affine component equal to 0. As

$\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 1 \\ 0 & 2 & 0 & -1 \\ \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ \frac{2}{\sqrt{2}} \\ 0 \end{pmatrix}$, the vector must be $(x, y, z) = (0, 4, \frac{2}{\sqrt{2}})$.

If we changed the value of $\mathbf{p}$, only the location of the point would change, because changing the destination of the origin only affects the affine part of the transformation and not the linear part.